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Section: 02

Ans. NO. 01(a)

Predicting house price(t) based on living area(x_1),
#bedrooms(x_2) with linear regression model:

$$\text{Model: } y = \theta_1 x_1 + \theta_2 x_2 + \theta_0$$

$$\text{Cost function: } J(\theta) = \frac{1}{2n} \sum (y(i) - t(i))^2, \text{ with } n=5 \text{ samples}$$

Given, Initial weights and bias, $\theta_1 = \theta_2 = \theta_0 = 0$

Learning rate, $\alpha = 0.000001$

With Gradient descent,

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{5} \sum (y - t)$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{5} \sum (y - t) x_1$$

$$\frac{\partial J}{\partial \theta_2} = \frac{1}{5} \sum (y - t) x_2$$

Iteration 1

$$y_{\text{pred}} = 0 \times x_1 + 0 = 0$$

$$y_{\text{pred}} = 0 \times x_1 + 0 \times x_2 + 0 = 0$$

Computing gradients:

$$\text{For } \theta_0, \theta_0(\text{new}) = \theta_0(\text{old}) - \alpha \frac{\partial J}{\partial \theta_0}$$

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{5} [(0-400) + (0-330) + (0-369) + (0-232) + (0-516)]$$

$$= -369.4$$

$$\theta_0 = 0 - [0.000001 \times (-369.4)]$$

$$= 0.0003694$$

For θ_1 , $\theta_1(\text{new}) = \theta_1(\text{old}) - \alpha \frac{\partial J}{\partial \theta_1}$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{5} [(-400 \times 2104) + (-330 \times 1600) + (-369 \times 2400) + (-232 \times 1416) + (-516 \times 3000)]$$

$$= -826342.4$$

$$\theta_1 = 0 - [0.000001 \times (-826342.4)]$$

$$= 0.8263424$$

For θ_2 , $\theta_2(\text{new}) = \theta_2(\text{old}) - \alpha \frac{\partial J}{\partial \theta_2}$

$$\frac{\partial J}{\partial \theta_2} = \frac{1}{5} [(-400 \times 3) + (-330 \times 3) + (-369 \times 3) + (-232 \times 2) + (-516 \times 4)]$$

$$= -1165$$

$$\theta_2 = 0 - [0.000001 \times (-1165)]$$

$$= 0.001165$$

$$\text{Cost} = \frac{1}{2 \times 5} [(-400)^2 + (-330)^2 + (-369)^2 + (-232)^2 + (-516)^2] = 72519.1$$

for, iteration 2:

$$y = 0.8263424x_1 + 0.001165x_2 + 0.0003694$$

$$\text{for } (2104, 3) \Rightarrow y = 1738.63, (y-t) = 1738.63 - 400 = 1338.63$$

$$\text{for } (1600, 3) \Rightarrow y = 1322.15, (y-t) = 1322.15 - 330 = 992.15$$

$$\text{for } (2400, 3) \Rightarrow y = 1983.23, (y-t) = 1983.23 - 369 = 1614.23$$

$$\text{for } (1446, 2) \Rightarrow y = 1170.10, (y-t) = 1170.10 - 232 = 938.10$$

$$\text{for } (3000, 4) \Rightarrow y = 2479.03, (y-t) = 2479.03 - 516 = 1963.03$$

$$\text{Cost} = \frac{1}{10} [(1338.63)^2 + (992.15)^2 + (1614.23)^2 + (938.10)^2 + (1963.03)^2]$$

$$= 1011554.88$$

Cost went up higher, so need to re-adjust the parameters through gradient descent

$$\text{for, } \theta_0(\text{new}) = \theta_0(\text{old}) - \alpha \frac{\partial J}{\partial \theta_0}$$

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{5} [1338.63 + 992.15 + 1614.23 + \cancel{938.10} + 1963.03]$$

$$= 1369.23$$

$$\theta_0 = 0.0003694 - (0.000001 \times 1369.23) = 0.001000$$

$$\text{For } \theta_1(\text{new}) = \theta_1(\text{old}) - \alpha \frac{\partial \bar{J}}{\partial \theta_1}$$

$$\begin{aligned} \frac{\partial \bar{J}}{\partial \theta_1} &= \frac{1}{5} [(1338.63 \times 2104) + (992.15 \times 1600) \\ &\quad + (1614.23 \times 2400) + (938.10 \times 1416) \\ &\quad + (1963.03 \times 3000)] \\ &= 3099101.88 \end{aligned}$$

$$\begin{aligned} \theta_1 &= 0.8263424 - (0.000001 \times 3099101.88) \\ &= -2.27276 \end{aligned}$$

$$\text{For } \theta_2(\text{new}) = \theta_2(\text{old}) - \alpha \frac{\partial \bar{J}}{\partial \theta_2}$$

$$\begin{aligned} \frac{\partial \bar{J}}{\partial \theta_2} &= \frac{1}{5} [(1338.63 \times 3) + (992.15 \times 3) + (1614.23 \times 3) \\ &\quad + (938.1 \times 2) + (1963.03 \times 4)] \\ &= 4312.67 \end{aligned}$$

$$\begin{aligned} \theta_2 &= 0.001165 - (0.000001 \times 4312.67) \\ &= 0.003148 \end{aligned}$$

$$\therefore y = -2.27276x_1 + 0.003148x_2 + 0.001000$$

(b)

from (a), $y = -2.27276x_1 + 0.003148x_2 + 0.001000$

features, # bedrooms, $x_2 = 4$

Living area, $x_1 = 220$

for $x_1 = 220$, $x_2 = 4$;

$$\begin{aligned} y_{\text{pred}} &= (-2.27276 \times 220) + (0.003148 \times 4) \\ &\quad + 0.001000 \\ &= -499.998 \end{aligned}$$

Meaning, two iterations wasn't useful to give a valid prediction of house price.

Ans. NO. 02

(a)

Logistic regression for feature $\rightarrow x_1$: living area (feet²)
 x_2 : # bedrooms

Predict house price classes: low, medium, high

Label encoding: low $\rightarrow 01 \Rightarrow z_1$

medium $\rightarrow 2$

high $\rightarrow 3$

for,

$$\text{low} = 1, z_1 = \theta_{11}x_1 + \theta_{12}x_2 + \theta_{10}$$

$$\text{medium} = 2, z_2 = \theta_{21}x_1 + \theta_{22}x_2 + \theta_{20}$$

$$\text{high} = 3, z_3 = \theta_{31}x_1 + \theta_{32}x_2 + \theta_{30}$$

Initially, $\theta_{11} = \theta_{12} = \theta_{10} = \theta_{21} = \theta_{22} = \theta_{20} = \theta_{31} = \theta_{32} = \theta_{30} = 0$

learning rate, $\alpha = 0.000001$

Iteration 1

Initially for all $\theta_{ij} = 0$, $z_1 = z_2 = z_3 = 0$ for all 3 classes

$$\text{Softmax activation: } P(y=1|x) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} = \frac{1}{3}$$

$$P(y=2|x) = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}} = \frac{1}{3}$$

$$\text{Similarly, } P(y=3|x) = \frac{1}{3}$$

Predicted class: $y = \arg \max [P(y=1|x), P(y=2|x), P(y=3|x)]$

Cross-entropy loss:

$$\begin{aligned} J &= -\frac{1}{5} \times [\log(\frac{1}{3}) + \log(\frac{1}{3}) + \log(\frac{1}{3}) + \log(\frac{1}{3}) \\ &\quad + \log(\frac{1}{3})] \\ &= -\log(\frac{1}{3}) \\ &= 1.0986 \end{aligned}$$

Gradient descent

Class-1 (low): $\frac{\partial J}{\partial \theta_{10}} = \frac{1}{5} \left[\left(\frac{1}{3} - 0\right) + \left(\frac{1}{3} - 1\right) + \left(\frac{1}{3} - 0\right) + \left(\frac{1}{3} - 1\right) + \left(\frac{1}{3} - 0\right) \right]$

$$= -0.0667$$

$$\begin{aligned} \theta_{10} &= \theta_{10}(\text{old}) - \alpha \frac{\partial J}{\partial \theta_{10}} \\ &= 0 - 0.000001 \times (-0.0667) \\ &= 0.0000000667 = 6.67 \times 10^{-8} \end{aligned}$$

$$\begin{aligned} \frac{\partial J}{\partial \theta_{11}} &= \frac{1}{5} \left[\frac{1}{3} \times 2104 + \left(\frac{-2}{3} \times 1600\right) + \frac{1}{3} \times 2400 \right. \\ &\quad \left. + \frac{-2}{3} \times 1416 + \frac{1}{3} \times 3000 \right] \\ &= 98.133 \end{aligned}$$

$$\theta_{11} = \theta_{11}(\text{old}) - \alpha \frac{\partial J}{\partial \theta_{11}} = -9.81 \times 10^{-5}$$

$$\frac{\partial \bar{J}}{\partial \Theta_{12}} = \frac{1}{5} \left[\frac{1}{3} \times 3 + \left(\frac{-2}{3} \times 3 \right) + \frac{1}{3} \times 3 + \left(\frac{-2}{3} \times 2 \right) + \left(\frac{1}{3} \times 4 \right) \right]$$

$$= 0$$

$$\therefore \Theta_{12} = 0$$

Case (2) - Medium

$$\Theta_{20} = \Theta_{20} - \alpha \frac{\partial \bar{J}}{\partial \Theta_{20}}$$

$$= 0 - \alpha \left[\frac{1}{5} \left[\left(\frac{1}{3} - 1 \right) + \frac{1}{3} + \left(\frac{1}{3} - 1 \right) + \frac{1}{3} + \frac{1}{3} \right] \right]$$

$$= 6.67 \times 10^{-8}$$

$$\Theta_{21} = \Theta_{21} - \alpha \frac{\partial \bar{J}}{\partial \Theta_{21}}$$

$$= \Theta_{21} - \alpha \left[\frac{1}{5} \times \left[\frac{-2}{3} \times 2104 + \frac{1600}{3} - \frac{2}{3} \times 2400 + \frac{1416}{3} + \frac{2000}{3} \right] \right]$$

$$= 1.9947 \times 10^{-4}$$

$$\Theta_{22} = \Theta_{22} - \alpha \frac{\partial \bar{J}}{\partial \Theta_{22}}$$

$$= 0 - \alpha \times \frac{1}{5} \left[\frac{-2 \times 3}{3} + \frac{3}{3} - \frac{2 \times 3}{3} + \frac{2}{3} + \frac{4}{3} \right]$$

$$= 2 \times 10^{-7}$$

Case (23) High

~~$$\theta_{30} = 0 - \alpha \times \frac{1}{5} \left[4 \times \frac{1}{3} + \left(\frac{1}{3} - 1 \right) \right]$$~~

$$\theta_{30} = 0 - \alpha \times \frac{1}{5} \left[-\frac{4}{3} + \left(\frac{1}{3} - 1 \right) \right]$$

$$= -1.33 \times 10^{-7}$$

$$\theta_{31} = 0 - \alpha \times \frac{1}{5} \left[\frac{2104}{3} + \frac{1600}{3} + \frac{2400}{3} + \frac{1416}{3} - \frac{2 \times 3000}{3} \right]$$

$$= -1.0133 \times 10^{-4}$$

$$\theta_{32} = 0 - \alpha \times \frac{1}{5} \left[1 + 1 + 1 + \frac{2}{3} - \frac{2 \times 4}{3} \right]$$

$$= -2 \times 10^{-7}$$

Iteration 2

For (2104, 3) $\Rightarrow z_1 = -0.2065 \Rightarrow p_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} = \frac{0.813}{2.3349} = 0.2588$

$z_2 = 0.4197 \Rightarrow p_2 = 0.4841$

$z_3 = -0.2132 \Rightarrow p_3 = 0.2571$

~~$\therefore z_1 + z_2 + z_3 \Rightarrow$~~

$$e^{z_1} + e^{z_2} + e^{z_3} = 3.1429$$

For (1600, 3) $\Rightarrow z_1 = -0.157, p_1 = \frac{e^{-0.157}}{3.081} = 0.2774$

$z_2 = 0.3191, p_2 = 0.4466$

$z_3 = -0.1621, p_3 = 0.2760$

$$(2400, 3) \Rightarrow z_1 = -0.2355, p_1 = \frac{e^{-0.2355}}{3.188} = 0.2478$$

$$z_2 = 0.4787, p_2 = 0.5062$$

$$z_3 = -0.2432, p_3 = 0.2459$$

$$(1416, 2) \Rightarrow z_1 = -0.1390, p_1 = \frac{e^{-z_1}}{3.063} = 0.2841$$

$$z_2 = 0.2824, p_2 = 0.433$$

$$z_3 = -0.143, p_3 = 0.2828$$

$$(3000, 4) \Rightarrow z_1 = -0.2944, p_1 = \frac{e^{-0.2944}}{3.302} = 0.2256$$

$$z_2 = 0.5984, p_2 = 0.5509$$

$$z_3 = -0.304, p_3 = 0.2235$$

Cross-entropy loss:

$$J(\theta) = -\frac{1}{5} [\log(0.4841) + \log(0.2774) + \log(0.5062) \\ + \log(0.2841) + \log(0.2235)]$$

$$= \cancel{1.082} = \cancel{1.079}$$

$$= 1.089$$

Gradient descent

Class(1) → Low

$$\frac{\partial J}{\partial \theta_{10}} = \frac{1}{5} [0.2588 + (0.2774 - 1) + 0.2478 + (0.2891 - 1) + 0.2256]$$

$$= -0.1412$$

$$\theta_{10} = 2.079 \times 10^{-7}$$

$$\frac{\partial J}{\partial \theta_{11}} = \frac{1}{5} [0.2588 \times 2104 - 0.7226 \times 1600 + 0.2478 \times 2400 + (-0.7159 \times 1416) + 0.2256 \times 3000]$$

$$= -70.768$$

$$\theta_{11} = -2.74 \times 10^{-5}$$

$$\theta_{12} = \theta_{10} - \alpha \times \frac{1}{5} [0.2354] = 2.355 \times 10^{-7}$$

Case(2) - Medium

$$\frac{\partial J}{\partial \theta_{20}} = \frac{1}{5} [(0.4891 - 1) + 0.4466 + (0.5062 - 1) + 0.433 + 0.5509]$$

$$= \frac{1}{5} [-0.5159 + 0.4466 - 0.4938 + 0.433 + 0.5509]$$

$$= 0.0842$$

$$\therefore \theta_{20} = 6.67 \times 10^{-8} - \alpha(0.0842) \\ = -1.746 \times 10^{-8}$$

$$\theta_{21} = 1.9947 \times 10^{-4} - \alpha(141.96) \\ = 5.75 \times 10^{-5}$$

$$\theta_{22} = 2 \times 10^{-7} - \alpha(0.276) \\ = -7.6 \times 10^{-8}$$

Case(3)-High

$$\frac{\partial J}{\partial \theta_{30}} = \frac{1}{5} [0.2571 + 0.2760 + 0.2459 + 0.2828 \\ + (0.2235 - 1)]$$

$$= \frac{1}{5} [0.2571 + 0.276 + 0.2459 + 0.2828 \\ - 0.7765]$$

$$= 0.0571$$

$$\therefore \theta_{30} = (-1.33 \times 10^{-7}) - \alpha(0.0571) = -1.90 \times 10^{-7}$$

$$\theta_{31} = (-1.0133 \times 10^{-4}) - \alpha(71.271) = -3.005 \times 10^{-5}$$

$$\theta_{32} = (-2 \times 10^{-7}) - \alpha(-0.0407) = -1.59 \times 10^{-7}$$

(b)

A house with $x_1 = 2200$ sq. feet, $x_2 = 4$ bedrooms

$$z_1 = (-2.74 \times 10^5 \times 2200) + (2.355 \times 10^7 \times 4) + 2.079 \times 10^7 \\ = -0.0602$$

$$z_2 = (5.75 \times 10^5 \times 2200) + (-7.6 \times 10^8 \times 4) + (-1.746 \times 10^8) \\ = \text{0.2} \text{ 0.126}$$

$$z_3 = (-3.005 \times 10^5 \times 2200) + (-1.59 \times 10^7 \times 4) + (-1.9 \times 10^7) \\ = -0.066$$

$$P(\text{y=Low}) \quad P(y=1|x) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} = \frac{0.9415}{3.01199} = 0.3126$$

$$P(y=2|x) = \frac{e^{0.126}}{3.01199} = 0.3766$$

$$P(y=3|x) = \frac{e^{-0.066}}{3.01199} = 0.3108$$

$$\text{Argmax} [P(y=1|x), P(y=2|x), P(y=3|x)] \\ = 0.3766 = P(y=2|x)$$

$\therefore$ The price level is medium

Ans. NO. 03

(a)

MSE/MAE assume that the relationship between linear input and output is ~~is~~ continuous; which is a rare case in Classification problems. In these type of problems, it leads to inefficient or misleading gradient upgrades. If a model predicts value greater than 1 or smaller than 0, MSE ends up cutting down valuable optimization.

Hence, cross-entropy loss is used for classification task. This ensures to penalize the wrong predictions much heavily. It helps to find the minimum cost function and with parameter update like gradient descent. Especially, as both Cross-entropy and classification work with probabilistic models. It treats the classification as probability estimation problem ranged from 0 to 1, as per the definition. It also ensures a converging minima through gradient descent.

(b)

Overfitting	Underfitting
① Model learns training data too well, including noise & random fluctuations.	① When model is too simple to capture underlying patterns in data.
② Low bias, high variance	② High bias, low variance
③ Having too little data to train and learn the patterns, too many parameters or too long training.	③ Not training model long enough, using too few iterations for updating parameters with gradient descent.
④ Training error very low.	④ Training error high.
⑤ Need regularization, simpler model, early stopping to fix	⑤ Increasing ML complexity, adding features, reducing regularization helps.

(c)

Regularization is a technique to stop models from becoming overly complex. It penalizes large weights, coefficients, so the models can generalize better for unseen data.

↳ L_1 Regularization (Lasso):

Adds the sum of the absolute values of the model coefficients to the loss function as penalty.

$$L_1 \text{ Penalty} = \sum_{j=0}^n | \theta_j |$$

Pushes some weights to zero, performing auto feature selection and giving sparse models. Useful when suspecting many features are irrelevant.

L₂ regularization (Ridge):

Adds the sum of the squares of the weights to the cost function as penalty.

$$L_2 \text{ Penalty} = \frac{1}{2} \sum_{j=0}^n \theta_j^2$$

Shrinks weights closer to zero, but not absolute zero. It leads to some what relevancy of all features.

(d)

Types of gradient descents:

① Batch gradient descent: Gradient computed using entire training set. The parameters are only updated after summing up and averaging all errors across the dataset.

- - Stable and accurate convergence but slow
- for large datasets as it need to go through the whole set for every iteration.

2) Stochastic Gradient Descent

Uses one training example at a time before performing ~~gradient descent~~ updating the parameters

- Faster and more memory efficient for large datasets. Can escape local minima due to its noisy updates.
- Less stable ~~du~~ because convergence process can be noisy. The cost fluctuates more

(3) Minibatch gradient descent

Somewhere between batch and stochastic gradient descent. Computes GD on a small part of the dataset and updates the parameter.

It takes stability of Batch GD and efficiency of SGD. Can perform matrix operation. But each mini-batch (small subset of dataset) requires tuning.